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Dear Editorial Committee,

We are submitting the revised version of the research paper titled “Towards Individual Channel Classifiers and Ensemble Methods for Enhanced Motor Imagery BCI Performance” submitted for SIPAIM2025. This version presents the requested improvements, following the reviewers and editor suggestions. Likewise, in the next page, we detailed each raised point and how we addressed all of them. We hope that this version will fulfill your expectations and being considered for publication.

Best regards,

Jaime A. Riascos-Salas, Hernán Villota, Marta Molina

Reviewer #1

1. The study only used data from 10 randomly chosen participants out of 109 available. This small sample size may not be representative of the broader population, potentially limiting the generalizability of the findings.

Answer: Yes, the reviewer is right in highlighting the small sample size used for our study. As this is an initial exploratory study, we decided to start with a small sample because, for the training, it is computationally expensive and time-consuming. Therefore, we plan to execute, considering the achieved results, a full dataset study in future experiments.

2. Training individual classifiers for each EEG channel (256 in total) could be computationally intensive, especially for real-time applications. The study doesn't explicitly address how this approach would scale for practical, real-time BCI systems.

Answer: We measured (not reported by space restriction) the prediction time in online setup, where each individual classifier takes around 3.5 ms. Training the full individual classifier during training takes time, but for online, you do need to re-train the model, they will work in prediction mode, where it takes around 3.5ms to predict a label for each time window. Therefore, our approach would work in real-time BCI systems because first you are not using all the 256 models and secondly, you are using them in prediction mode, without any further training. We improved the IV Classification section to reduce any confusion, with the following:

“To prepare the models for real-time simulation, all final classifiers were trained on the complete training dataset (runs 6 and 10) and stored to use them in prediction mode for future tests on a completely unseen EEG run (run 14) using a sliding window of 0.5 seconds that moved with a step size of 0.1 seconds.”

3. Selecting only the 10 best-performing individual classifiers per subject might lead to overfitting, especially given the limited number of trials per class (14-15). The study doesn't discuss measures taken to prevent overfitting in this selection process.

Answer: The selection of the 10 best-performing individual classifiers per subject cannot lead to any overfitting because all of them are trained with the same amount of data. Therefore, there is no relationship between the number of classifiers and overfitting in them. We chose the 10 best as representative number because, after this number, the accuracy started to get the same accuracy until it got a big difference later after 30-40 classifiers. We will run in future studies an optimization algorithm to choose the optimal number of classifiers per subject.

4. The simulated online BCI scenario shows significant variability in classifier performance over time, but the study doesn't propose a method to address this instability in real-time applications.

Answer: Yes, this is one of the weaknesses of the proposed method that should be addressed with further investigations. While it is usual that the variability among subjects (between sessions and within subjects), we perceived that, compared with FBCSP and mRMR methods, our individual channel classifiers behave with greater variability. Although it should be noticed that we are reporting individual performance in our approach vs average performance in both methods (FBCS, mRMR) so Figure 4 should be read carefully. We modified the caption and expanded the Figure's explanation to clarify this. Nevertheless, to go further about online behavior, it is crucial to understand the same dynamics of the EEG signal over time, where subjects can either present greater performance at the beginning of the epoch or at the end. This is not a new phenomenon in BCI, as users could be affected by fatigue, cognitive stress, discomfort, and any other human factor during the interaction.

"It is important to notice that the individual classifiers are coming for each subject, while the FBCSP and mRMR methods use the average among subjects; this partially explains the variability within the epoch among the individual classifiers.

... The variability in our individual approach highlights the dynamic nature of BCI signals and the challenge of maintaining stable, high accuracy in an online setting, where subjects can either present greater performance at the beginning of the epoch or at the end. This is not a new phenomenon in BCI, as users could be affected by fatigue, cognitive stress, discomfort, and any other human factor during the interaction...."

5. While the study presents accuracy improvements, it doesn't provide statistical tests to confirm if these improvements are significant, especially given the small sample size.

Answer: Significant tests were not carried out during this initial study. We will consider it in a major experiment where we will include a bigger sample size.

Reviewer #2

1. It is not entirely clear why the 10 best classifiers per subject were selected. I suggest clarifying this point in the manuscript (why is this number relevant?).

Answer: We chose the 10 best as representative number because, after this number, the accuracy started to get the same accuracy until it got a big difference later after 30-40 classifiers. We will run in future studies an optimization algorithm to choose the optimal number of classifiers per subject. This was included in the paper.

"Finally, we selected the 10 best classifiers as a representative number, but future optimization studies should be carried out to determine the ideal number of classifiers per subject."

2. Considering the use of adaptive BCI systems, it would be valuable to discuss whether other types of classifiers could be integrated into this approach, particularly those that might further improve accuracy.

Answer: It is not clear the connection between adaptive BCI systems and the use of different types of classifiers. If we understood correctly, it would be the use of different types of algorithms together to investigate their performance. As it would be always an endless study, we explored the most representative methods used in literature individually as this is an initial approach where we wanted to check its feasibility. Adaptive BCI would be a matter of further studies once we could validate the individual channel classification over different scenarios, setups and datasets.

3. You state: "Intriguingly, regions around the parietal lobule are also achieving high accuracy rates (blue regions), with some occipital areas (only for MLP) being particularly active. KNN, SVM, and MLP classifiers were notably effective in these regions." This is an interesting finding, and I suggest expanding the interpretation. Specifically: Why might the parietal lobule show higher accuracy rates? What could explain the efficiency of these classifiers in these anatomical regions?

Answer: While it is a matter of further investigation to check whether this finding is presented in several datasets or not, we could preliminarily argue that parietal areas also play an important role in BCI because among the functions delegated for this brain region, authors mention sensory processing, attention, and spatial and motor control in visuomotor action (Culham & Valyear, 2006). Most of the BCI research focuses on motor cortex where C and CP channels are included but here we can perceive that P channels also is a valid option for BCI classification. One potential explanation for the achieved accuracy could be the attention and motor control reflected in the signals from these channels are easier to identify and classify. It was included in the paper.

Reviewer #3

1. The paper presents an analysis of an ensemble-based methodology for classification in MI-BCIs. The idea is interesting and the results are promising.

Towards Individual Channel Classifiers and Ensemble Methods for Enhanced Motor Imagery BCI Performance

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***Abstract*—This study introduces a novel approach to enhance Motor Imagery Brain-Computer Interface (MI-BCI) classification by training individual channel classifiers and integrating them with a stacking ensemble. Our method was benchmarked against established techniques, including Filter Bank Common Spatial Patterns (FBCSP) and Minimum Redundancy Maximum Relevance (mRMR). While a full pool of 256 individual channel classifiers initially showed suboptimal performance, refining the approach by selecting only the 10 best-performing individual classifiers per subject yielded substantial accuracy improvements of over 20%. This curated selection consistently outperformed FBCSP and, for some subjects, even mRMR. The findings suggest that focusing on a personalized, high-performing subset of channels can effectively reduce inter- and intra-subject variability, offering a robust alternative to traditional optimization-based channel selection. This framework holds promise for developing more adaptable and effective BCI systems.**

I. Introduction

Brain-Computer Interfaces (BCIs) that utilize motor imagery (MI) are a popular method for individuals to control external devices using their brain signals [1]. This is done by mentally rehearsing movements, such as moving a hand or a foot, without actually performing them. This mental practice changes brain activity in the sensorimotor cortex [2], specifically in the mu and beta frequency bands, which can be measured with an electroencephalogram (EEG) at electrodes like C3, Cz, and C4. These changes are known as Event-Related Synchronization (ERS) and Desynchronization (ERD) and are often referred to as sensorimotor rhythms (SMRs) [3]. Applications range from controlling prostheses and wheelchairs to virtual reality and video games [4].

However, using EEG-based BCIs faces several challenges [5]. The electrode caps can be uncomfortable, and movement of the user's head or body can create noise in the signal, called artifacts. Additionally, the brain signals can vary a lot between different people and even in the same person on different days, which makes it difficult to consistently and accurately classify the user's intended commands [6]. A key part of the BCI process is converting the raw EEG data into useful commands. This involves two main steps: feature extraction and classification. EEG data is often collected from many channels (sometimes over 100), which can be noisy and contain redundant information. This large amount of data increases the time and computational power needed to process it and can lead to less accurate models. To address this, an important research effort has focused on finding the most important and minimal set of EEG channels [4]. This channel selection process is crucial for reducing complexity, improving accuracy, and making BCIs more efficient and user-friendly.

The usual approaches for reducing the EEG dimension are often adapted from feature selection algorithms, source representation, and filtering principles [4]. Common Spatial Pattern (CSP) is a popular spatial filtering method for MI-EEG analysis that maximizes the variance between different motor imagery classes, making them easier to distinguish [7]. A major improvement, Filter Bank Common Spatial Pattern (FBCSP), applies this principle across multiple frequency bands, often leading to better classification accuracy by identifying the most relevant spectral information for a given user and task [8]. For example, one study showed that FBCSP-LDA achieved an average accuracy of 73.23% in a dynamic MI experiment, highlighting its effectiveness in real-world scenarios [8]. Another powerful method, Minimum Redundancy Maximum Relevance (mRMR), is a feature selection technique that selects a subset of features that are highly relevant to the classification task while minimizing redundant information among them [9]. This approach can be applied to features from various domains—spatial, temporal, and spectral—and has been shown to achieve strong performance. For instance, Javidan et al. reported an average accuracy of 79.1% using a reduced feature set on a standard BCI dataset [10]. Furthermore, to address the challenge of multi-electrode setups, researchers have explored Genetic Algorithms (GA) to optimize the selection of a minimal set of electrodes. Soler et al. used a Non-dominated Sorting Genetic Algorithm (NSGA-II) to find electrode subsets that minimize both the number of electrodes and the localization error of the brain source [11]. They suggested that even with as few as six electrodes, it could achieve an accuracy equal to or better than systems with over 200 channels for single-source cases, proving that strategic, optimized electrode placement is more critical than a high-density, unoptimized setup [11]. These techniques collectively represent noteworthy advancements in creating robust, efficient, and user-friendly MI-BCIs by refining how we process brain signals and select the most informative data.

The traditional approach to improving MI-BCIs focuses on finding a single best group of channels to use for all people and all situations. However, because everyone's brain activity is different, the best channels for a specific MI task can change from person to person and even from one session to the next. This paper proposes a different strategy: instead of using a single set of channels, it suggests training a separate, individual classifier for each EEG channel. The outputs from these individual classifiers are then combined into an ensemble, a classification technique that leverages the collective ”wisdom” of multiple models. This novel method aims to see if localizing the classification to each channel and then pooling the results can be a more personalized, computationally efficient, and reliable solution for MI-BCI, especially for real-time use. Therefore, our experiment specifically focuses on refining the feature extraction and classification stages from the typical BCI pipeline [5].

II. Materials and Methods

A. EEG Motor Movement/Imagery Dataset Physionet

Using the publicly available EEG Motor Movement/Imagery Dataset from PhysioNet [12], we selected data from 10 participants, randomly chosen, out of 109 total. This dataset was recorded with 64 electrodes and captured brain activity during both imagined and executed movements. For our specific experiment, we focused on runs 6 and 10 from each participant, which involved imagining opening and closing both fists or both feet. We used this data for training our models, while reserving run 14—a separate run of the same imagined movements—as a dedicated test set to evaluate performance over time.

B. Preprocessing

Raw EEG data was put through a standard pre-processing pipeline using the MNE-Python library. First, we loaded the data and selected only the EEG channels, applying a standard ’10-05’ montage for consistent electrode placement. We did not apply any initial bandpass filtering or Common Average Reference (CAR) to the data to keep the integrity of the raw information for the individual channel classification approach. Then, the continuous data was segmented into discrete epochs, or trials, locked to the start of the motor imagery cues. For offline analysis, epochs were from 0.5s to 3.5s after the cue, while a wider window (-1.0s to 4.0s) was used for simulating online classification. Finally, we removed any epochs with artifacts and ensured the remaining number of epochs for each motor imagery class was equal to prevent bias. After this process, four subjects had a complete set of 15 trials per class, while the others had 14 trials per class.

III. Feature Extraction Scenarios

A. Band Power

To extract frequency-domain features from the EEG time series, we calculated the average power in five distinct frequency bands for each of the 64 channels: Theta (4-7 Hz), Alpha (8-12 Hz), Low Beta (13-20 Hz), High Beta (20-30 Hz), and Gamma (30-45 Hz). Welch's method was used for these calculations. This process yielded five features per channel, creating a total initial pool of 320 features (64 channels × 5 bands), which were used to train individual channel classifiers and as the source for the mRMR feature selection process.

B. Filter Bank Common Spatial Patterns (FBCSP)

FBCSP is a powerful technique that combines frequency filtering and spatial filtering to enhance the differences between motor imagery classes [7]. We applied five different 4 Hz wide bandpass filters, spanning from 8 Hz to 28 Hz, to the EEG data. For each of these five filtered bands, we applied the Common Spatial Patterns (CSP) algorithm with Ledoit-Wolf regularization to generate four components. The logarithm of the variance of these components was then extracted as features, resulting in a total of 20 features (5 bands × 4 components).

C. Minimum Redundancy Maximum Relevance (mRMR)

mRMR is a feature selection method designed to identify the best subset of features by maximizing their relevance to the classification task while minimizing their redundancy [9]. We applied mRMR to the initial pool of 320 Band Power features (64 channels × 5 frequency bands). Using mutual information as the selection criterion, mRMR was used to select the top 20 most informative features from this large feature space.

IV. Classification

We assessed four common BCI classifiers [13]: Support Vector Machine (SVM), Multilayer Perceptron (MLP), K-Nearest Neighbors (KNN), and Linear Discriminant Analysis (LDA) [13]. We used these as base classifiers and optimized their hyperparameters using 5-fold stratified cross-validation. These classifiers were trained in three distinct ways. First, they were trained separately on the 20 features selected by the mRMR method and also on the 20 FBCSP features. Second, for our novel approach, we trained a separate classifier for each channel using the Band Power features, resulting in 256 individual classifiers. We then chose the 10 best-performing individual classifiers to be the foundation for two ensemble methods: Majority Vote, which simply takes the most common prediction, and Stacking, which uses the predictions of the 10 individual classifiers as input for a ”meta-classifier” (an MLP) to make a final decision. The entire process, from feature extraction to classification, was managed using a Sci-kit learn Pipeline. To prepare the models for real-time simulation, all final classifiers were trained on the complete training dataset (runs 6 and 10) and stored to use them in prediction mode for future tests on a completely unseen EEG run (run 14) using a sliding window of 0.5 seconds that moved with a step size of 0.1 seconds. For each model, we measured classification accuracy. The entire experiment was conducted on a Dell Precision 3680 Tower with an Intel i9-14900K processor, 32GB RAM, and a 1TB SSD.

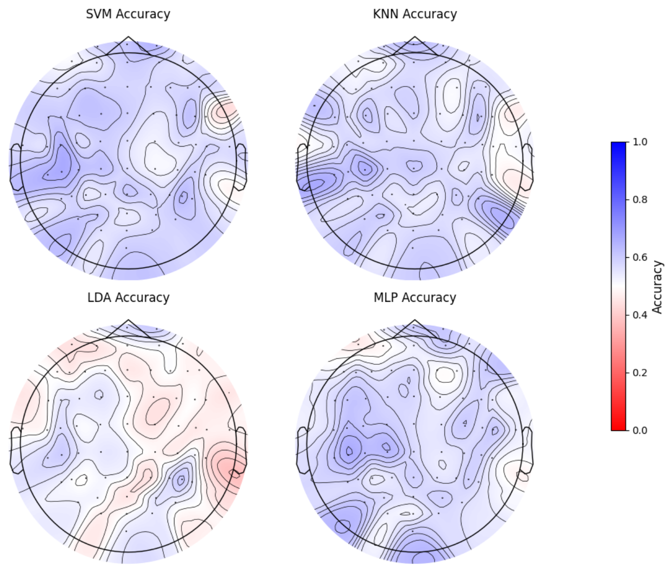

Fig. 1. The topographical distribution of individual channel accuracies, averaged across all subjects. Blue areas show accuracy above the chance level of 0.5, white the chance level, and red below the chance level. Accuracies were calculated from the training data using stratified 5-fold cross-validation.

TABLE I
THE MEAN AND STANDARD ERROR CLASSIFICATION ACCURACIES FOR INDIVIDUAL CHANNEL CLASSIFIERS, ENSEMBLE METHODS, AND THE 10 BEST-PERFORMING CLASSIFIERS VERSUS FBCSP AND MRMR APPROACHES. ACCURACIES RANK FROM HIGH TO LOW PERFORMANCE.

Individual channel approach			FBCSP and mRMR approach	
Subject	Mean Accuracy Full Pool Classifiers	Mean Accuracy 10-Best Classifiers	Classifier Type	Mean Accuracy
35	0.675 (0.094)	0.940 (0.033)	FBCSP MLP	0.727 (0.070)
62	0.616 (0.087)	0.898 (0.063)	FBCSP SVM	0.697 (0.068)
40	0.598 (0.089)	0.839 (0.077)	FBCSP KNN	0.695 (0.062)
58	0.573 (0.085)	0.807 (0.102)	FBCSP LDA	0.649 (0.087)
73	0.550 (0.088)	0.778 (0.091)	Mean	0.692 (0.072)
99	0.534 (0.082)	0.749 (0.068)	mRMR KNN	0.830 (0.074)
23	0.516 (0.087)	0.744 (0.086)	mRMR MLP	0.830 (0.080)
06	0.502 (0.102)	0.730 (0.108)	mRMR SVM	0.808 (0.079)
94	0.501 (0.084)	0.748 (0.067)	mRMR LDA	0.665 (0.067)
32	0.483 (0.096)	0.695 (0.077)	Mean	0.783 (0.075)
Mean	0.555 (0.089)	0.792 (0.077)		
Stacking	-	0.729 (0.086)		
Majority Vote	-	0.600 (0.000)		

V. RESULTS AND DISCUSSION

Our study evaluates the performance of our proposed individual and ensemble classifiers against established methods (FBCSP and mRMR). We present our findings in three stages: i) the topographical distribution of the mean accuracies for all individual channel (Figure 1), and the general performance of the full pool and top-performing 10 classifiers (Table I) against the established method; ii) 10-best classifiers analysis through the individual Channel performance (Figure 2) and the Channel-Classifier performance (Figure 3); and iii) performance analysis in a simulated real-time classification (Figure 4).

Initial analysis of all individual classifiers indicates that electrodes over the left temporal and central cortices—the brain's motor cortex— consistently achieve higher classification accuracy than chance level. Intriguingly, regions around the Parietal lobule are also achieving high accuracy rates (blue regions), with some occipital areas (only for MLP) being particularly active. KNN, SVM, and MLP classifiers were notably effective in these regions. LDA has several brain regions under the chance level. Most of the MI-BCI research focuses on Motor Cortex, findings here suggest that Parietal channels also is a valid option for MI-BCI classification. One potential explanation for the achieved accuracy could be that the patterns produced by this brain region, related to sensory processing, attention, and spatial and motor control in visuomotor action [14], are more tangible to recognize and classify.

Table I presents a comparative analysis of the individual channel approach versus the FBCSP and mRMR methods in terms of classification accuracy. Across all subjects, the overall mean accuracy for the full pool of individual channel classifiers was 0.555±0.089. This shows a high degree of inter-subject variability, with individual mean accuracies ranging from a maximum of 0.675±0.094 to a minimum of 0.483±0.096, a value close to chance level (0.5). When we consider the subset of the 10 best-performing individual channel classifiers per subject, a substantial improvement is observed. The overall mean accuracy rises to 0.792±0.077, marking an approximately 42% increase over the full pool. All subjects showed substantial gains, with mean accuracies ranging from a minimum of 0.695±0.077 to a maximum of 0.940±0.033. These results highlight the effectiveness of selecting the top-performing classifiers to boost accuracy. The ensemble methods, Stacking and Majority Vote, also performed well on this subset, achieving accuracies of 0.729±0.086 and 0.600±0.000, respectively.

The FBCSP and mRMR methods demonstrated strong performance. The average accuracy for FBCSP was 0.692±0.072, while the mRMR method showed an even higher average of 0.783±0.075. Notably, the mRMR with either KNN or MLP classifiers achieved the highest single accuracies, both at 0.830. Comparing the different approaches reveals some key insights. The 10-best individual channel classifier approach surpasses the average performance of the FBCSP method and is comparable to the average performance of the mRMR method. Specifically, the mean accuracy of the 10-best classifiers is higher than that of all individual FBCSP classifiers, and it is similar to the highest-performing mRMR classifiers. This suggests that a carefully selected subset of individual channel classifiers can be as effective, or even more effective, than more complex feature extraction methods like FBCSP and mRMR.

The aggregated classification accuracies for the 10 best-performing channels are depicted in Figure 2. The mean accuracy of these top 10 channels ranges from a maximum of approximately 0.95 to a minimum of approximately 0.80 (mean = 0.852±0.071). This indicates a very high performance level for the selected channels, where CP6, Fp2, CP4, CP3, F7, and C6 outperformed the established methodologies and ensemble methods. The superior performance of channels in

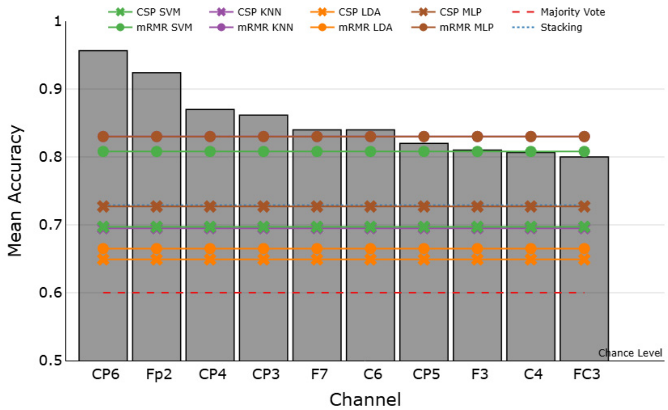

Fig. 2. Mean accuracy per channel for 10-best individual channel classifiers. Lines represent feature extraction methods and ensemble methods. Chance level is at 0.5.

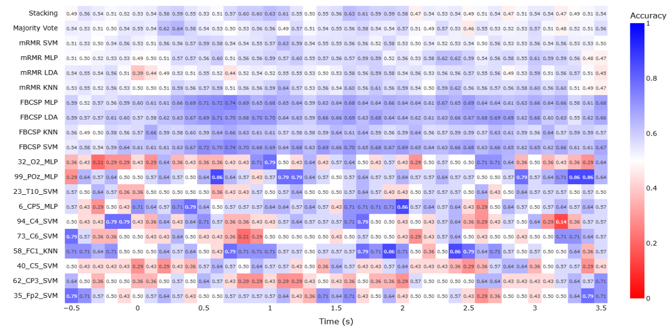

Fig. 4. Classification rates over time for unseen data. The X-axis represents the time from cue onset (0s), while the Y-axis lists the best individual channel classifier per subject, the average among subjects of each base classifier for each extraction method, and ensemble classifiers. The sliding window duration is 0.5 s, with a step size of 0.1 s.

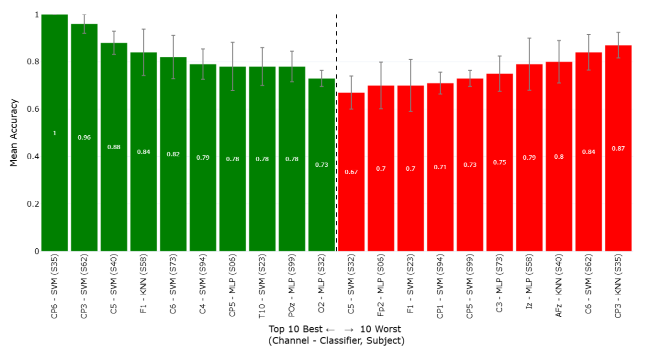

Fig. 3. Performance distribution of the 10 best and 10 worst classifiers per subject from the 100-classifier pool, derived from the original 256 individual channel classifiers. Groups are demarcated by a vertical dotted line and color-coding.

the central, parietal, and frontal regions is particularly notable, confirming their importance for classification tasks in this context.

Figure 3 shows a comparative analysis of the 10 best and 10 worst-performing classifiers selected from the consolidated pool of 100 top classifiers (10 best per subject). The 10 best classifiers (green bars) exhibit higher accuracies, ranging from 0.73 to 1.00 (mean = 0.836±0.062), with the highest achieved by the CP6 - SVM classifier for subject 35. In contrast, the 10 worst classifiers (red bars) from the same consolidated pool show a lower range of accuracies, from 0.67 to 0.87 (mean = 0.756±0.076). Interestingly, SVM is among the most used classifiers (6 out of 10 from the best group, and 5 out of 10 from the worst group), and LDA did not reach any high performance. While these classifiers still perform well above chance level (0.5), the performance gap between the two groups underscores the value of selecting a specific subset of top-performing classifiers. This selection process effectively filters out less effective models, leading to a pool of highly specialized and accurate classifiers.

Figure 4 displays an over-time classification accuracy heatmap of various classifiers and methodologies, simulating an online BCI scenario. The results show great variability in performance across both classifiers and time windows, as expected in online BCI. It is important to notice that the individual classifiers are coming for each subject, while the FBCSP and mRMR methods use the average among subjects; this partially explains the variability within the epoch among the individual classifiers. Notably, the FBCSP-based classifiers and ensemble methods generally maintain accuracies around or above the chance level throughout the epoch. Specifically, the FBCSP MLP, FBCSP LDA, and FBCSP SVM classifiers tend to show higher accuracies, with peak performance often occurring between 1.5 and 2.5 seconds. The Stacking ensemble also performs well, consistently staying above chance. In contrast, several individual channel classifiers exhibit greater performance fluctuations. Some, like the 99_P02_MLP classifier, show a high accuracy of 0.86 towards the end of the epoch, while others, such as 32_O2_MLP, perform well below chance level during early and mid-time windows. The mRMR-based classifiers also show varying performance, with the mRMR KNN and mRMR MLP achieving periods of above-chance accuracy. The variability in our individual approach highlights the dynamic nature of BCI signals and the challenge of maintaining stable, high accuracy in an online setting, where subjects can either present greater performance at the beginning of the epoch or at the end. This is not a new phenomenon in BCI, as users could be affected by fatigue, cognitive stress, discomfort, and any other human factor during the interaction. Finally, the heatmap effectively visualizes these complex temporal dynamics, showing that while some classifiers can achieve high peak performance, others struggle to maintain consistent accuracy over time.

VI. Conclusions

By leveraging a curated subset of the top-performing individual channel classifiers, this study successfully addresses the accuracy limitations of a full-channel approach in MI-BCI. The selection of the 10 best classifiers per subject yielded a notable performance increase of over 20%, resulting in accuracies that consistently outperformed established FBCSP methods and, for some subjects, even mRMR and ensemble classifiers. Although this innovative and personalized approach, which utilizes the most effective individual classifiers per subject, presented better average accuracies, it failed to mitigate the variability in accuracy over time during the online prediction. Nevertheless, the findings suggest that this method potentially serves as a valuable alternative to traditional channel selection, offering a framework for developing more robust and adaptive BCI systems. Still, additional research is needed to validate its use. The presented results also support the hypothesis that asymmetric, non-standard electrode configurations can provide superior performance by focusing on specific, high-contributing brain regions. Finally, we selected the 10 best classifiers as a representative number, but future optimization studies should be carried out to determine the ideal number of classifiers per subject.

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